

Film Scheduling Strategy Optimization: The Introduction of Squeeze and Stopping Point Model for Movie Box Office Prediction

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Abstract. The 2019 novel coronavirus pneumonia (COVID-19) epidemic has caused a great impact on the Chinese film market. In recent years, the Chinese box office market has rebounded, but it has not recovered to the pre-epidemic level. Based on the box office data of 3791 Chinese movies from 2016 to 2023 provided by Maoyan Film, this paper proposes the concept of Film market Depression, identifies stopping points through stopping point analysis, and establishes a multiple regression model to explore the impact of competition. This study found that most stopping points occurred within two weeks of a movie's release, with new releases within three days likely to cause stopping points to reduce box office revenue, but high quality and popular movies suffered less revenue loss. This study innovatively analyzes the impact of dynamic competition on movie box office from a stopping point perspective. The research results show that the film market squeeze has a negative impact on the box office of the film, and the rational arrangement of films to reduce the negative impact of squeeze is helpful for the film to achieve better box office performance. The research results are helpful to optimize the arrangement strategy and promote the development of the Chinese film market.

Keywords: Box office forecast; Stopping point model; Marketing; Management statistics.

1. Introduction

Since 2019, China's box-office has continued to decline due to the COVID-19 epidemic. Figure 1 Data released by Maoyan Research Institute shows that in 2020 and 2022, the average annual total box-office of China's film market is 32.580 billion yuan, much lower than the 64.266 billion yuan in 2019. In 2023, with the end of the epidemic, China's film market rebounded, and the annual total box office recovered to 54.915 billion yuan, but it still has not recovered to the level of 55.911 billion yuan in 2017 before the epidemic, and there is a gap between it and the peak in 2019. [1]

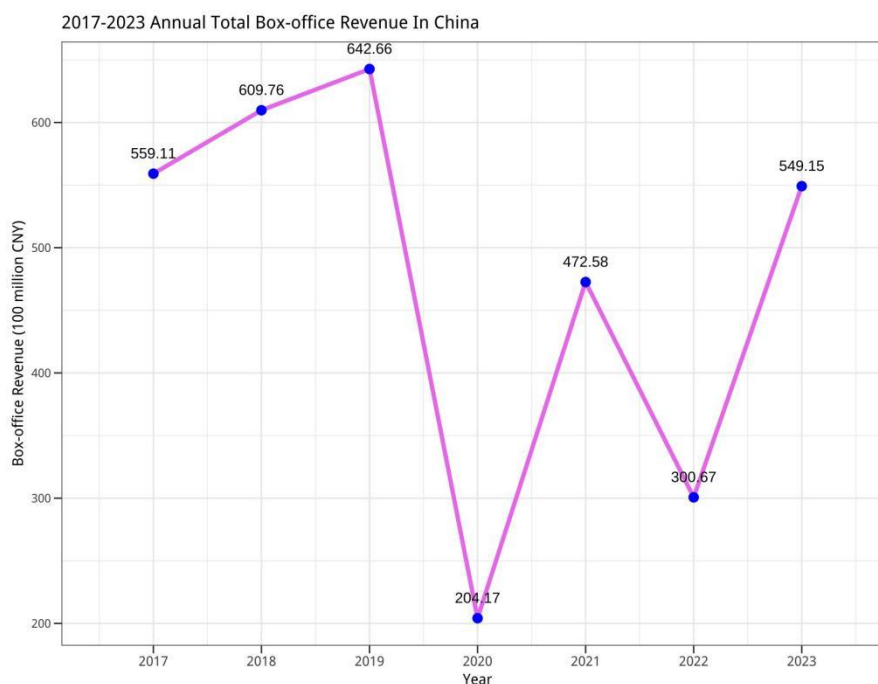


Fig. 1 Total annual box office in China 2017-2023

But in the larger context, an important phenomenon that is different from the pre-pandemic period deserves our attention: the decline in the share of imported films in the market. According to Lighthouse data [2], only 78 imported films will be released in 2023, compared with 124 in 2019. At the same time, in terms of box office, the box office of domestic films in 2023 reached 46.05 billion yuan, becoming the highest box office of domestic films since 2019. In terms of the annual total box office of imported films, although compared with 3.324 billion in 2020, 7.331.5 billion in 2021 and 4.556 billion in 2022, the annual total box office of 8.91 billion in 2023 shows an obvious upward trend, however compared with 23.091 billion in 2019, the annual box office of imported films in 2023 shows an obvious downward trend. Showing a more obvious downward trend. On the whole, the box office proportion of domestic films will remain at 83 in 2023. For imported films, there are certain advantages in the box office appeal and the consumer purchase promotion force to promote the audience's consumption, and it is expected to achieve further development under the help of appropriate strategies.

On the whole, the film market is changing rapidly, and the new scheduling strategy considering the time factor is particularly important in such a market environment. This paper defines the negative impact of high-earning blockbusters on the box-office revenue of other films in the same period as squeeze, which we call Film market Depression in English. At the same time, it further discusses the impact of competition on the box office and introduces stopping point analysis in order to further explore the impact of competition on the box office and find new scheduling strategies to provide support for the recovery of the Chinese film market.

The data used in this article was provided by Maoyan Film Company. The dataset covers information related to 3,791 films released in China from 2016 to 2023, including aspects such as daily box office revenue, film quality and film content. Descriptive statistics for all variables are shown in Table 2. Box office receipts (Box-Office Revenue), BOR, are box office receipts measured in 2023 real value after adjusting for the effect of inflation. The variable ranges from about 0 to 5.775 billion yuan, with an average of 99.03 million yuan. Since the distribution of box office receipts is skewed, we use a logarithmic transformation: $\ln BOR = \ln(BOR + 1)$. The range of the dependent variable after the transformation is 0 to 9.76, with an average of 5.86.

2. Organization of the Text

2.1 Movie Box office Data

According to Irson and Grotfelty (2012) [3], the Star Power (SP) of a film has a significant impact on the final box office of the film. Based on suggestions from industry experts, we defined star power as a discrete variable of 05 based on the number of films in which a star participated and its box office performance. An actor who has participated in five or more films and whose average box office is more than 1 million yuan is defined as level 5 influence. An actor who has acted in five or more films with an average box office of less than 1 million is defined as level 4. An actor who has acted in between two and four films with an average box office of more than 1 million is a level 3 influence; An actor who has acted in between two and four films with an average box office of less than 1 million is a Level 2 influence; Those who have appeared in only one film are considered level 1. Those with no record of the actor in the database are defined as 0.

With the rapid development of Internet platforms, various online Internet platforms have gathered a large number of potential audiences, and the evaluation of a movie on the platform often reflects the quality of a movie. As one of the largest film ticketing and evaluation platforms in China, the data of Maoyan Professional Edition is extremely representative. We use the film to watch films (NumbersofPotentialAudience, NPA) and score (MovieAudienceRating, MAR) two indicators to measure the evaluation of the audience for movies. Among them, the number of wanted viewers refers to the number of users who express their desire to watch a movie on the platform before the movie is released. Generally speaking, the more films want to see, the higher the quality of promotion, the more potential audiences, and the higher the final box office of the film. The movie score is the rating that the audience gives to the movie after watching the movie, and the rating should be between 0 and 10 points.

Table 1. Descriptive statistics of variables related to movies

Variable	Min	Median	Max	Mean	Std.Dev.Unit
BOR	0	1.55 x 10 ²	5.775 x 10 ⁵	9.90 x 10 ³	3.87× 1,040,000 yuan
lnBOR	0	6.19	9.76	5.86	2.22
SP	0	3	5	2.69	1.73
NPA	0	3804	4468547	46958.26	151423
MAR	0	0	9.8	3.73	4.08

2.2 Construct the movie box office prediction model

2.2.1 Test of traditional model

We first build a movie box office prediction model from the perspective of movie content and quality, which serves as the basis for our subsequent analysis.

$$\ln \text{BOR}_i = \alpha_0 + \alpha_1 \times \text{Quality} + \alpha_2 \times \text{Content} + \epsilon_i \quad (1)$$

$\ln \text{BOR}_{i,t}$ in equation 6 is the real box office income of painting and calligraphy; Quality is a vector to measure the quality of the film, including three variables: the rating of the leading actor, the number of wanted to see, and the cat's eye score; Content is a vector measuring the content of a movie, which contains multiple dummy variables representing the genre of the movie.

We estimate the parameters of model (1) using the relevant data of 3,791 films released in the Chinese mainland from 2016 to 2023, and the results are shown in Model 1 of Table below.

Variable Estimate Std.Error t value Pr(> t)				
Variable	Estimate	Std.Error		

Star rating	4.166×10^{-1}	2.560×10^{-2}	16.273	$< 2 \times 10^{-16}$
Want to see the number	4.498×10^{-6}	2.243×10^{-7}	20.053	$< 2 \times 10^{-16}$
Cat-eye rating	4.789×10^{-1}	9.604×10^{-3}	49.868	$< 2 \times 10^{-16}$
Main Type action	$- 1.331 \times 10^0$	1.943×10^{-1}	6.849	8.76×10^{-12}
Main type Children	$- 2.286 \times 10^0$	3.297×10^{-1}	6.934	4.85×10^{-12}
Main Type Crime Mystery	$- 1.727 \times 10^0$	1.662×10^{-1}	10.392	$< 2 \times 10^{-16}$
Main type Period costume	$- 2.090 \times 10^0$	3.463×10^{-1}	6.037	1.74×10^{-9}
Main type Foreign War Historical Biographies	$- 1.641 \times 10^0$	3.064×10^{-1}	5.355	9.12×10^{-8}
Main Genre documentary	$- 1.507 \times 10^0$	1.951×10^{-1}	7.721	1.50×10^{-14}
Main Type Family	$- 2.280 \times 10^0$	1.861×10^{-1}	12.252	$< 2 \times 10^{-16}$
Main Type Thriller Horror	$- 1.233 \times 10^0$	1.666×10^{-1}	7.399	1.71×10^{-13}
Main type drama	$- 2.143 \times 10^0$	1.492×10^{-1}	14.357	$< 2 \times 10^{-16}$
Main genre Sci-fi	-7.131×10^{-1}	2.337×10^{-1}	3.051	0.002299
Main Type Fantasy Adventure	$- 1.026 \times 10^0$	4.028×10^{-1}	2.547	0.010899
Main Type Youthful Love	$- 2.195 \times 10^0$	1.519×10^{-1}	14.455	$< 2 \times 10^{-16}$
Main type Wuxia	$- 2.363 \times 10^0$	5.093×10^{-1}	4.639	3.63×10^{-6}
Main genre Comedy	$- 2.127 \times 10^0$	1.618×10^{-1}	13.143	$< 2 \times 10^{-16}$
Main type opera	$- 3.015 \times 10^0$	3.671×10^{-1}	8.212	3.03×10^{-16}
Main type Music Song and dance	-9.599×10^{-1}	3.553×10^{-1}	- 2.702	0.006935
Main Type Exercise	$- 2.321 \times 10^0$	3.225×10^{-1}	- 7.198	7.47×10^{-13}
Main type disaster	$- 1.615 \times 10^0$	5.678×10^{-1}	- 2.845	0.004462
Main Type Chinese historical biography main theme	-8.105×10^{-1}	2.279×10^{-1}	- 3.556	0.000381
Main Type China War	$- 1.337 \times 10^0$	2.351×10^{-1}	- 5.687	1.40×10^{-8}

Table 2. Regression coefficients of traditional models

Our empirical results prove that movie quality can significantly affect movie box office. To be specific, starring rating, which represents the pre-production input and production budget of a film, has an important positive impact on the final box office, and this impact has both economic and statistical significance. In addition, both the number of wants to see and the cat's eye score show a significant positive correlation with the final box office performance, which indicates that the degree of attraction of the film to the audience and the audience's evaluation have a positive impact on the box office.

In terms of film content, there are also significant differences in the final box office performance of different types of films. Taking animation films as the benchmark group, we find that animation films, historical biography themes, action films, crime and suspense films are more likely to win the favor of consumers, while drama, science fiction and drama films still have a gap in box office performance. These findings are consistent with the research results of the existing literature.

In addition, according to LiliKang et al. [4], movies with low, medium and high input may have different characteristics in the prediction model. He suggests that star rating be used as a proxy variable for film input, and regression is performed in segments. Based on the star rating, we build the following classification model:

$$\begin{cases} \ln BOR_i = \alpha_0 + \alpha_1 \times Quality + \alpha_2 \times Content + \epsilon_i & StarPower = 0,1 \\ \ln BOR_i = \alpha_0 + \alpha_1 \times Quality + \alpha_2 \times Content + \epsilon_i & StarPower = 2,3. \\ \ln BOR_i = \alpha_0 + \alpha_1 \times Quality + \alpha_2 \times Content + \epsilon_i & StarPower = 4,5 \end{cases} \quad (2)$$

The comparison shows a high consistency. Based on the adjusted R square of the above model and the P-values of various parameters, we believe that the overall regression model can better fit the box office in this movie data, which provides a basis for our subsequent analysis.

2.2.2 Unsqueezed box office prediction mode

There are 35 films in our data that are clearly labeled as not being squeezed in their three weeks of releasing, and the box office of these films reflects the true box office in the absence of squeezing. Below, we bring these into model 6 above to get the unsqueezed box office prediction model, which results in the table below.

Table 3. Regression results for unsqueezed movies

Variables	Estimate	Std.Error	tvalue	Pr(> t)
Starring rating	4.166 x 10 ⁻¹	2.560 x 10 ⁻²	16.273	< 2 x 10 ⁻¹⁶

Using the squeezed box office prediction model, we can calculate the predicted real box office (under unsqueezed conditions) of the squeezed movie and the box office loss due to the squeeze. The figure below shows this result.

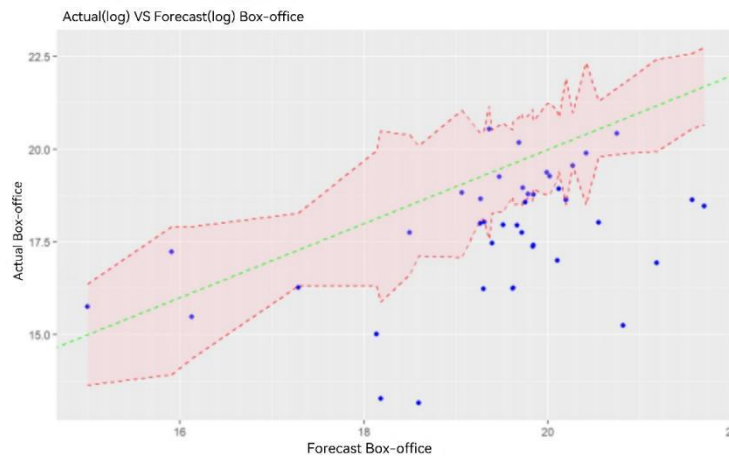


Fig. 2 Scatter plot of actual and predicted box office of squeezed films

The vertical axis shows the actual box office achieved by these films, the horizontal axis shows the predicted box office calculated by the model, and the red area is the confidence interval for the 95% confidence level of the predicted model. Most of the films fell below the line with a slope of 1, meaning that films clearly marked as squeezed did indeed underperform at the box office as expected. Some 43.9 % of the points fell outside the 95 %confidence interval, meaning that the box office losses suffered by these films were significant. These results all show that the phenomenon of competition and squeeze in the film market is real, and it is necessary to conduct further analysis of this phenomenon.

2.2.3 Breaking point analysis of extrusion

In order to better analyze the film market depression factors, we first need to identify the day on which the break point at the box office occurred. In a time series, a break point is the point at which there is a clear change in the forward and backward trend. Consider a classic time series model that includes stopping points:

$$X_t = \mu_t^0 + \epsilon_t, \quad t = 1, 2, \dots, n \quad (3)$$

In this paper, we carry out stopping point detection . For any two positive integers $0 \leq k_1 \leq k_2 \leq n$ we first define:

$$\bar{X}_{k_1, k_2} = (k_2 - k_1)^{-1} \sum_{t=k_1+1}^{k_2} X_t \quad (4)$$

$$S_n(k_1, k_2) = \sum_{t=k_1+1}^{k_2} (X_t - \bar{X}_{k_1, k_2})^2 \quad (5)$$

If there is a stopping point $k_1 < k < k_2$, define the residual sum of squares as:

$$L(k; k_1, k_2) = S_n(k_1, k) + S_n(k, k_2) \quad (6)$$

$$\hat{k} = \operatorname{argmin}_{k=1,2,\dots,n} L(k; k_1, k_2) \quad (7)$$

Using the stopping point analysis method described above, we label the extrusion stopping points in R language for 41 movies that are clearly labeled as being squeezed. Figure 3 shows the stopping point locations for some movies, and it can be found that our stopping point analysis results are in line with our definition of stopping points for box office data.

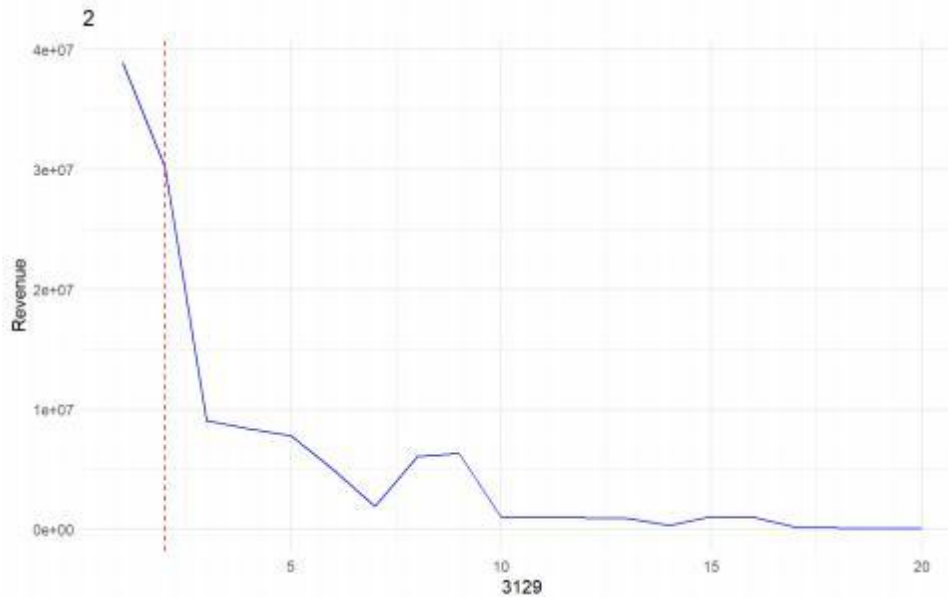


Fig. 3 Example of stopping point analysis

Table 3 shows the results of the stopping point analysis for all squeezed movies. The first column shows the name of the movie, the second column shows the days after the extrusion, and the third column shows the new movie released three days before the extrusion point, which is used as the extrusion movie to facilitate our subsequent analysis. From the table, we can find that most of the extrusion break points occurred between 1 and 14 days after the release of the movie, that is, within two weeks after the release of the movie. This period of time should arouse the attention of film producers.

Table 3. Stopping point analysis results

Movie Serial number	Number of days the stopping point occurred	Serial number of the movie released 3 days before the break point
2685	11	3766
3129	2	213,358,412,487,877,196 0,230 8,333 6,341 9
2041	13	2712
935	6	648,857,114 3,179 2,314 4,320 5,336 8,340 1,374 0
1793	2	87,211 1,223 0,278 2,288 5,340 2,352 2,371 6
3091	13	328 1
816	3	353,103 5,111 6,151 6,155 8,165 9,172 1,206 0, 286 5,371 9
2747	3	18,158,864,216 7,218 3
2026	2	784,106 4,120 2,152 5,166 0,245 5,323 5,325 1, 3373
1064	7	1055
3578	6	188 0,313 2,351 3
129	1	293,124 0,152 3,204 0,214 1,259 3,3753
2037	9	657,906,113 2,136 0,166 9,170 4,211 5,226 4,

		264 0,275 0,296 1,344 4,357 4
2220	9	70,230,474,635,874,109 4,133 6,142 5,194 0, 207 8,212 4,248 2,270 0
1053	2	612,801,145 3,150 5,165 5,174 2,197 1,197 6, 2427,347 2
1856	3	191,823,132 3,255 9,267 6,290 5,3697
93	3	2503,306 1,315 1
2762	2	413,709,223 7, 248 5 ,277 6 ,324 3 ,378 6
695	9	730, 3253
385	2	42, 251, 375, 610, 913, 1232, 1313,1626,2143,2613,2617,3731
1090	2	82, 156, 228, 439, 853, 1100, 1298, 1584, 1648, 2627, 3284, 3399, 3420
1603	2	1570, 1888, 2084, 2258, 2313, 2444, 2763, 3081, 3173, 3203, 3232, 3591
1345	5	2014, 2964
464	13	956
2771	3	40, 173, 286, 565, 1039, 1114, 1438, 2460, 3367, 3383, 3388, 3626
980	2	111, 351, 436, 1260, 1372, 1437, 1459, 1857, 1654,2513, 2707, 3577, 3641
3090	2	249, 617, 981, 1395, 1927
1509	2	313, 1632, 2340, 2727, 3068, 3685, 3717
3704	2	2303, 3066, 3316, 3562
3219	2	477, 975, 1419, 1526, 1958, 3435, 3473
1434	4	1026, 1384, 1531, 1722, 2872, 3349, 3356
1531	2	1026, 1384, 1722, 2872, 3349,
1384	2	1026, 1531, 1722, 2872, 3349, 3356
2352	5	947, 2125, 2808, 3561
965	10	192, 611, 1935, 2334, 3112, 3520
1402	15	429, 3166
1564	5	1850
324	2	462, 916, 2704, 3742

2.3 Extrusion factor analysis

In order to quantitatively analyze the box office loss of a movie due to the squeeze, we first define the box office loss rate:

$$\text{lossrate} = (\text{BOR}_{\text{predict}} - \text{BOR}_{\text{predict}}) / \text{BOR}_{\text{predict}} \quad (8)$$

The real box office is the result of the relative films competing and squeezing each other in the market, and the predicted box office is the result obtained by the prediction model in the ideal case without competition, so equation 13 is the expression of the loss rate caused by competitive squeezing.

Due to the omnipresent competition in the film market, any new film released within three days before the break point is likely to squeeze the original film. Therefore, we use AveragingModel's method to build a regression model.

$$\text{lossrate} = \alpha_0 + \alpha_1 \times \text{Quality}_j + \alpha_2 \times \text{Content}_j + \epsilon_i \quad (9)$$

Where lossrate denotes the rate of loss that film i will suffer due to extrusion; Quality_j represents the quality characteristics of squeezed films, including three variables: Maoyan rating, Starring rating and number of wants to see; Content_j represents the content characteristics of the squeezed

movie, which contains multiple dummy variables indicating the genre of the movie. According to Averaging Model's theory, the squeezed movie will randomly select one of the movies released three days before all the stopping points as the object of our analysis with a medium probability.

After 1000 experiments, the regression results obtained are shown in Table 4:

Table 4. Averaging Model regression coefficient table

Variables	Estimate	90% confidence interval
Maoyan rating	2.758×10^{-1}	$[-7.912 \times 10^{-1}, 1.879 \times 10^0]$
Starring rating	2.153×10^0	$[-3.630 \times 10^{-1}, 1.4977 \times 10^0]$
Number of wants to see	333×10^{-4}	$[-2.368 \times 10^{-5}, -5.900 \times 10^{-5}]$
Action Crime	1.049×10^{-1}	$[-2.341 \times 10^1, 3.019 \times 10^1]$
Drama	2.423×10^1	$[-9.631 \times 10^{-1}, 5.115 \times 10^1]$
Science Fiction Fantasy	2.752×10^1	$[1.345 \times 10^1, 4.781 \times 10^1]$
Comedy Family	5.254×10^0	$[-1.195 \times 10^1, 2.696 \times 10^1]$
War History Biographies	1.873×10^0	$[-2.970 \times 10^1, 3.360 \times 10^1]$

3. Literature References

3.1 The prediction model of mainstream film box office.

There are many factors affecting the movie box office. In order to better grasp the changes of the movie box office dynamically, BarryR.Litman and LindaS.Kohl (1989) [5] proposed the movie box office prediction model for the first time.

At present, the mainstream research on box office prediction tends to focus on the factors of the movie itself. KARNIOUCHINAEV (2010) [7] found that the popularity of stars in the opening week has a significant impact on the box office of a movie. Hot stars can attract the audience and cause a sensation, while the popularity of the movie itself is not only helpful for the opening week of a movie, but also promotes the box office during the whole release period of a movie. The study found that the role of stars in increasing the box office was much higher than previous studies thought. YEUNGCH et al. (2011) [2] analyzed the competition among films from the perspective of audiences, arguing that films with higher ratings and more reviews are more likely to attract more audiences and beat their competitors more easily. Pedro et al. (2016) [8] studied the determinants of box office revenue in the film industry and found that offensive content such as swear words or nudity may hinder a film from obtaining financial returns, while violent content seems to increase box office revenue. LiliKang et al. (2022) [4] investigated the box office performance of domestic films in China. It reveals that film neutrality and profanity in the Chinese market have a significant negative impact on box office revenue, and confirms the role of cultural values in the economic success of Chinese films. However, in big-budget films with stars, violence and gory content can boost box office. The mainstream model, based on pre-release data, does not take into account competition among films and how box office may change over time, and is not well adapted to the real box office market.

3.2 Introduction of competitive film box office prediction model.

The research on film market competition began in the last century. Krider and Weinberg (1998) [6] proposed the competition caused by the high seasonality of film business. The paper concludes by discussing the theoretical basis of "avoiding competition" in the broader context of product launch timing. Ainsliea et al. (2005) [9] examined previous methods of predicting box office and found that none of the previous studies introduced the competitive environment of the film. Therefore, he introduced the competitive factor of the film represented by the market share, and

found that the accuracy of the weekly movie box office prediction after the introduction of competition was improved compared with the traditional model.

By establishing different mathematical models, many scholars have made a more in-depth analysis of competition and its influencing factors on box office. CALANTONE et al. (2010) [10] explained the impact of competition on the box office by establishing an empirical model, and the result showed that all types of competition had a negative impact on the box office. He believed that films with earlier release date were more susceptible to the negative impact of competition than newly released films. EINAVL (2010) [11] established an empirical model to study the game of film release date, and found that the current film release schedule is too concentrated in large holidays and weekends, and if the distributor staggers the peak of film release, the box office revenue will increase. At the same time, some machine learning algorithms have also been introduced into the box office research. TaeguKim et al. (2015) [12] proposed a movie box office prediction method based on social network service (SNS) data and machine learning algorithm. The genetic algorithm is used to determine important input variables, and the prediction algorithms and explanatory variables used in the film field are comprehensively reviewed.

3.3 Film box office squeezing Film market Depression

There are many kinds of mutual influence between the box office of movies. There are many names for this phenomenon, competition, and the 80% effect, but few studies have defined this phenomenon systematically. In the film box office market, the negative impact of high-earning blockbusters on the box office revenue of other films in the same period is defined as squeeze, which we call Film market Depression in English for further discussion. The specific effects and mechanisms are as follows:

The research on film market competition began in the last century. Krider and Weinberg (1998) [6] proposed the competition caused by the high seasonality of film business. The paper concludes by discussing the theoretical basis of "avoiding competition" in the broader context of releasing time. Ansile et al. (2005)[13] found that when a film is released at the same time as other films of the same genre, it will have a negative impact on box office performance, reducing sales at the beginning, but in the long run, it will have a substitution effect, making the loss of sales less serious. According to the study of Calantone et al. (2010) [10], no matter what kind of competition will have a negative impact on the box office. A. Yizim Orhun, et al. (2015) [14] believe that competition will prompt theaters to show movies that are expected to achieve box office success more frequently, and at the same time stimulate theaters to adopt a new movie faster. There are various understandings of competition, but generally speaking, it is generally believed that competition is an inevitable and normal phenomenon in the film market.

The concept of the Blockbuster effect was coined by Frank and Cook(1996)[15] as the expression "winner takes all." The idea grew out of Rosen's (1981)[16] analysis of the economics of stardom: only a small percentage of artists or literary works are exceptionally successful. Later studies have confirmed the same in the movie market, with DeVany(2006)[17] arguing that "the top 20% of movies get 80% of the revenue." Blockbuster effect in the study of movie box office means that a small number of movies occupy the majority of the box office market, and when this part of movies is released, it may cause great losses to the box office of other movies that have been released.

EINAV,L(2010)[11] showed that the box office revenue would increase if distributors delayed some holiday release films by one or two weeks. Cabral et al. (2020) [18] study found that the relationship between film appeal and high-demand weekly releases is U-shaped: it declines at the level of lower overall appeal (niche films) and rises at the level of higher overall appeal (blockbusters). We can see that excellent film scheduling strategies can reduce the box office competition and the negative impact of Blockbuster effect on the box office to a certain extent.

To sum up, we define the negative impact of high-attraction blockbusters on the box office of low-attraction niche films in the same period of competition as film market depression, and further study on this basis.

3.4 Stopping point model and its application

A change point is a sudden change in time series data. This sudden change may represent a transition that occurs between states. Change Point Detection is an important method in time series model, which has a mature research framework at the theoretical level. Aminikhanghahi et al. [19] summarized various methods of stopping point analysis. They divided the stopping point analysis methods into supervised learning methods and unsupervised learning methods, and discussed the characteristics and application of various unsupervised learning methods. Table 1 summarizes the characteristics and applicability of all kinds of unsupervised learning methods.

There are also many scholars who have made new contributions to stopping point analysis. WangW,HeX et al. (2020)[20] introduced the weight bootstrap method on the basis of traditional stopping point analysis to obtain the confidence interval of stopping point estimation and improve the reliability of the analysis. Huang Miaoqi, Xia Zhiming (2023)[21] et al proposed the model averaging method and obtained its excellent theoretical properties based on the asymptotic theory -- the method has stronger stability, especially when the model is incorrectly set as a sudden change model.

Table 5. Features and applications of various unsupervised learning methods

Methods	Features	Application
Likelihood ratio method	Based on comparing the probability density of successive time intervals	Applies to one-dimensional time series
Subspace model method	Analyze the subspace of the time series	Suitable for multivariable time series and dynamic systems
Probabilistic methods	Use a probabilistic model for estimation	Explore the internal state of the time series
Kernel based methods	Map data to a high-dimensional feature space	The data is not linearly separable in the original space

The application of stopping point analysis in social science has been very extensive, in medicine, aerospace, finance, business, meteorology, entertainment and other fields have rich applications. Qianqianx (2018)[22] analyzed the annual data of food poisoning incidents in China and found no obvious downward trend break point, indicating that the effect of China's food safety protection policies is not obvious. Based on the physical conditions and technical data of football players within 90 minutes, CorbettDM,SweetingAJ(2019)[23] analyzed the characteristics of athletes' competitive level changing over time, which provided a basis for coaches' rotation decision. LynchC,MestelB(2019)[24] analyzed the situation of financial fraud from 2010 to 2018 based on the financial data of several countries in the European Union, and found that the number of financial fraud in various countries showed a breaking point of an upward trend near 2014, which may be related to the macroeconomic environment at that time and the change of accounting policies and other factors. In the field of film box office and film and television industry, we have not yet found any research on box office prediction from the perspective of stopping point analysis.

4. Summary

Based on the above research conclusions, we believe that movies can improve their box office performance in the following directions: (1) Consider competition and arrange the release time reasonably: movie distributors should fully consider the market competition environment and avoid releasing too many similar types of movies in the same period , so as to reduce the box office loss. (2) Improve film quality: Producers should pay more attention to improving the overall quality of films, including the story, production level and audience's word of mouth, so as to improve their

competitiveness, especially the films with less cost. (3) Improve marketing accuracy: distributors should formulate targeted marketing strategies according to the market performance of different types of films, give full play to the role of star effect and audience word-of-mouth, and maximize box office revenue. (4) Policy support: relevant government departments may consider improving relevant policies to support the production and distribution of diversified films, promote the healthy development of the film market and reduce vicious competition.

There are still some problems in our model, which can be further explored by considering other prediction models such as lasso regression in the future. Through the research and suggestions in this paper, we hope to provide useful references for film market participants, further promote the recovery of the film industry, and help the long-term healthy development of the film industry.

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